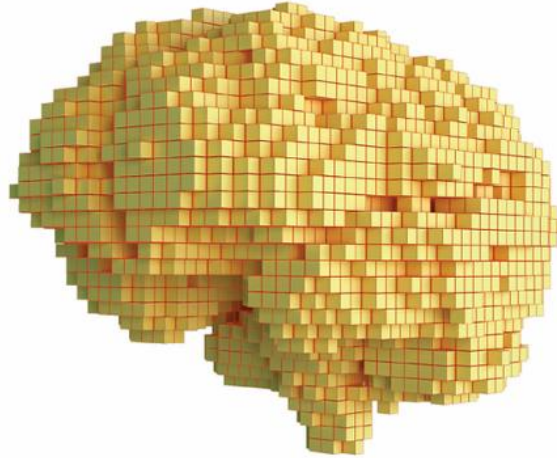


# Chapters 1 and 2 of Becoming a Data Head

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8/6/2026

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# **BECOMING A DATA HEAD**

HOW TO **THINK, SPEAK, AND  
UNDERSTAND DATA SCIENCE,  
STATISTICS, AND MACHINE LEARNING**

# The Core Story

1. What problem are we solving?
2. What data do we have?

# Why Data Projects Fail

- Starting with technology
- Unclear business value
- Wrong or incomplete data
- No clear end condition
- Results that nobody uses

# The Five Questions

1. Why is this problem important?
2. Who does it affect?
3. What if we do not have the right data?
4. When is the project over?
5. What if we do not like the results?

# 1. Why is this problem important?

- Why does this matter?
- What keeps the organization awake at night?
- Is this really a new problem?
- Has someone already solved it?
- What is the expected return on investment?
- What is the “size of the prize”?

## 2. Who will this problem affect?

The authors recommend pretending that the project has already been completed and asking:

1. Can we use the answer?
2. Whose work will change?

### 3. What if the right data does not exist?

- The necessary data does not exist.
- The available data measures the wrong thing.
- Important groups are missing.
- The data is too incomplete or unreliable.
- New data must be collected.
- The original problem must be narrowed or changed.

# 4. When is the project over?

- The final deliverable
- The success metric
- The expected business outcome
- The project deadline
- Whether ongoing maintenance will be needed
- Who will own the result after delivery

# 5. What if we do not like the results?

- A marketing campaign did not work.
- A popular product is not profitable.
- A leader's preferred strategy performs poorly.
- The data contradicts the original business assumption.

# Two Major Warning Signs



# 1. Methodology focus

- “We need artificial intelligence.”
- “Let’s use sentiment analysis.”
- “We should build a machine-learning model.”

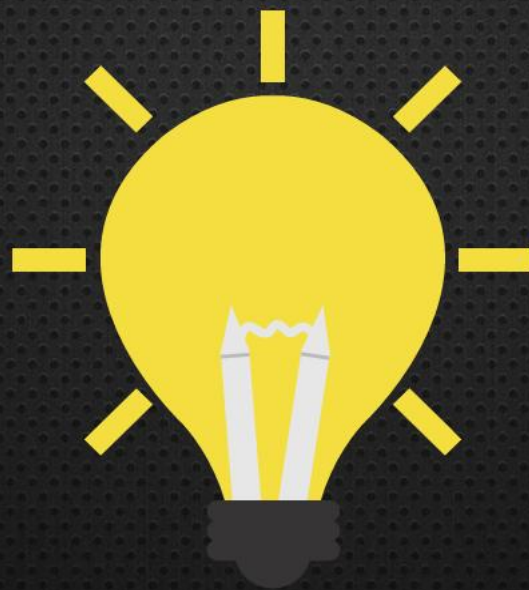
## 2. Deliverable focus

- “We need a dashboard.”
- “We need an app.”
- “We need a real-time reporting system.”



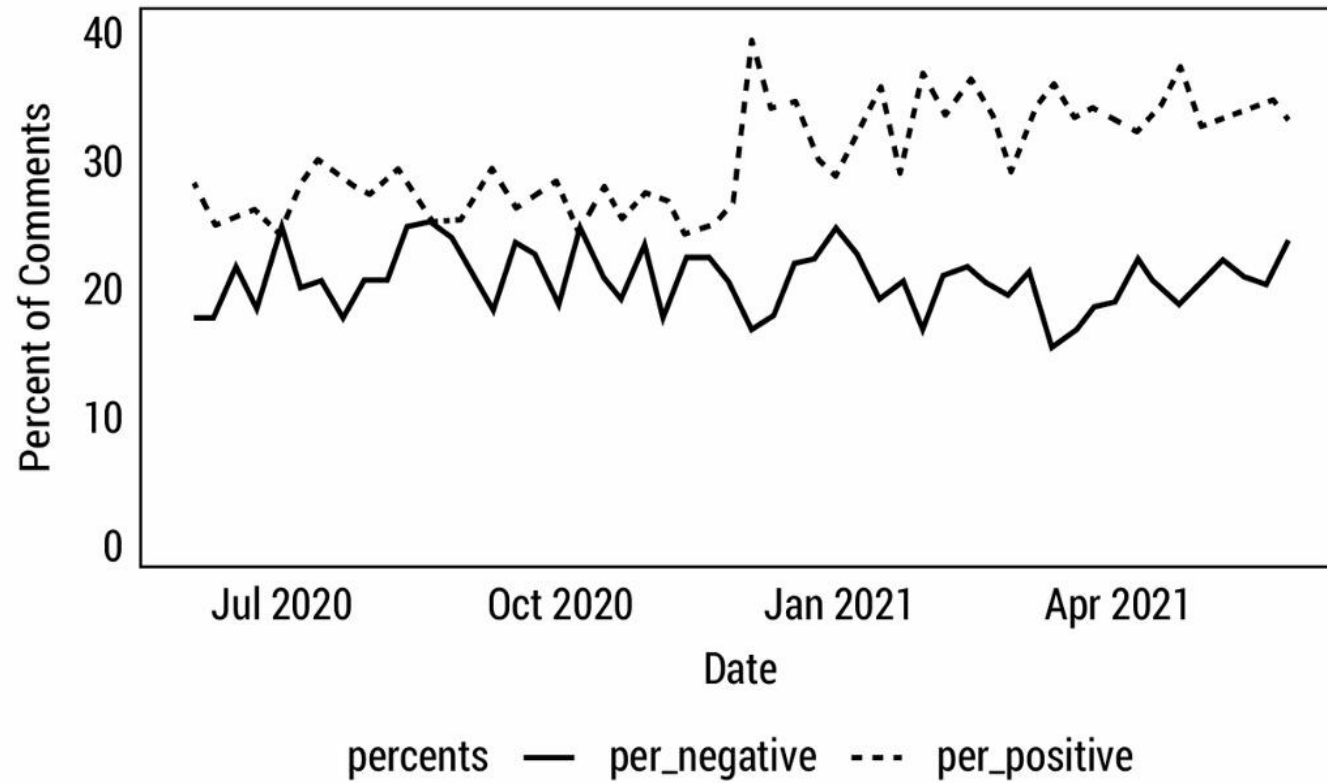
# Case Study: The Customer Perception Dashboard

We should use sentiment analysis



## Sentiment Analysis

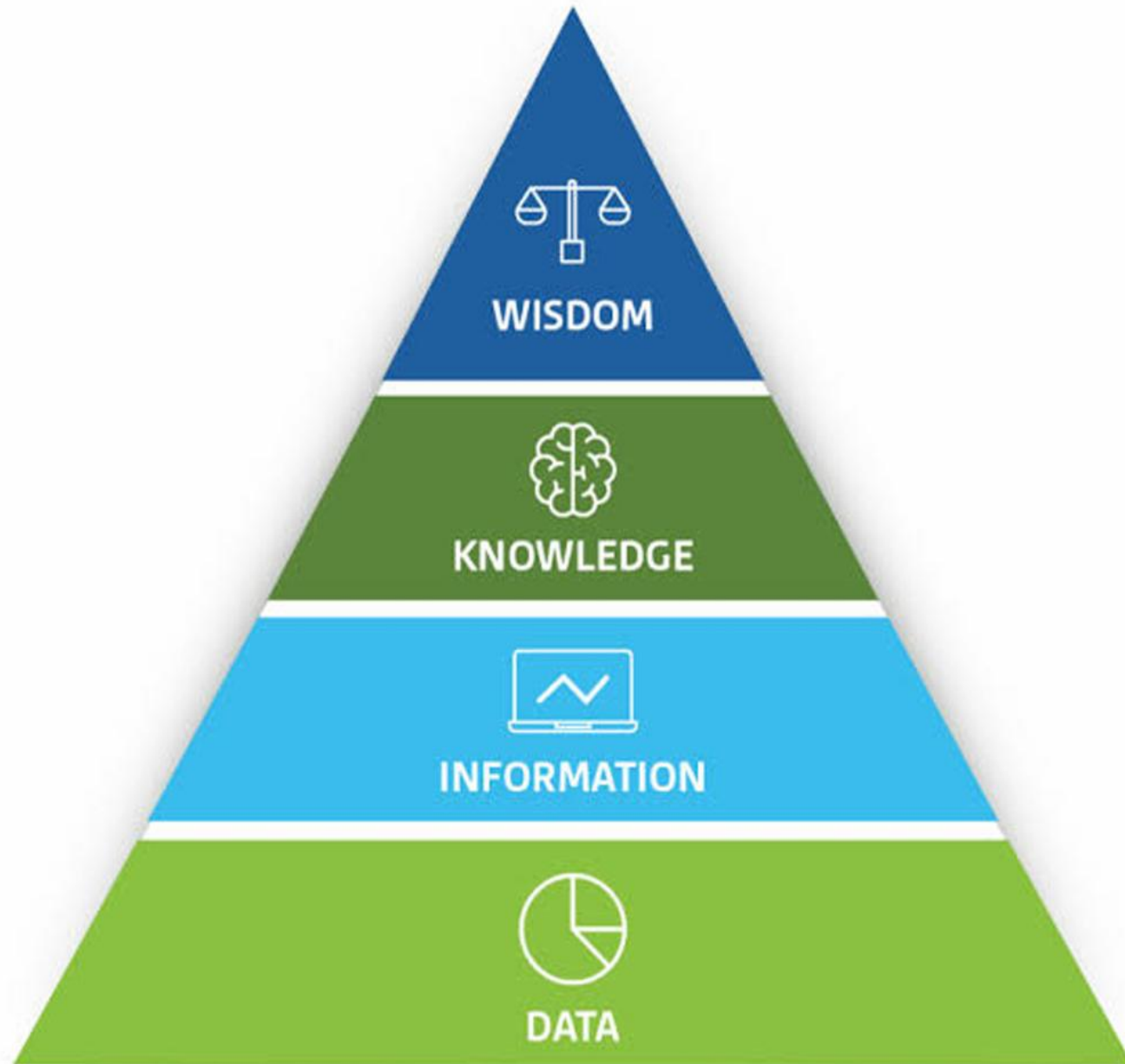
Last 12 Months



# The chapter cites workplace barriers reported by data scientists:

- Lack of a clear question: 30.4%
- Results not used by decision-makers: 24.3%
- Lack of domain-expert input: 19.6%
- Unrealistic expectations of project impact: 15.8%
- Difficulty integrating findings into decisions: 13.6%

# Chapter 2: What is data?



Rows may also be called:

- Observations
- Records
- Tuples
- Trials

Columns may also be called:

- Features
- Variables
- Fields
- Attributes
- Predictors

# Know Your Audience's Language

A statistician may say “variable,” while a machine-learning professional may say “feature.” A database professional may say “field.”

# Data Types

- Numeric data
- Categorical data

# Numeric data

## A. Continuous data:

Temperature

Height

Weight

Time

Distance

## B. discrete data:

Number of customers:

Number of cars

Number of complaints

Number of purchases

# Categorical data

## A. Ordered or ordinal data:

- Small, medium, large
- Poor, fair, good, excellent
- Satisfaction ratings from 1 to 10

## B. Unordered or nominal data:

- Online, television, print
- Yes, no
- Country
- Department
- Product type

# Structured and Unstructured Data

## Structured data:

- Spreadsheets
- Database tables
- Transaction records
- Customer lists

## Unstructured data:

- Written reviews
- Emails
- Photographs
- Videos
- Audio recordings
- Social-media posts

# How data is collected changes the claim

## OBSERVATIONAL

Records what happens

## EXPERIMENTAL

Tests what causes it

# Mean, Median and Mode

Data = 2, 4, 4, 4, 5, 7, 8, 9, 10

Mean: 15.89

Median: 5

Mode: 4

# Conclusion

**Problem first. Data second.**

The background is a dark, textured surface with a fine, repeating dot pattern. In the corners, there are abstract geometric elements: the top-left and bottom-right corners feature overlapping dark gray triangles with thin gold lines and clusters of small white dots; the top-right and bottom-left corners are decorated with flowing, wavy lines composed of many thin, light gray curves.

Thank You